FAULT TOLERANT CONTROL OF THE BOEING 747 SHORT-PERIOD MODE USING THE ADMISSIBLE MODEL MATCHING TECHNIQUE

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Abstract: The paper analyzes the different model matching approaches (Exact, Pseudo-Inverse Method / PIM, Modified Pseudo-Inverse Method / MPIM) for the fault tolerant control of the short-period mode of the longitudinal model of the Boeing 747 aircraft. To overcome the drawbacks of the PIM and MPIM approaches, the paper proposes a new philosophy for the model matching problem, namely to use a set of admissible models instead of trying to best approximate a single reference one. The application shows the benefits associated with using this approach. *Copyright © 2006 IFAC*

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1. INTRODUCTION

Fault tolerance (FT) has established itself as an important research area in the last decade (Blanke *et al.*, 2003), (Zhang and Jiang, 2003). It refers to the mechanism by which a given system is still able to exhibit given properties, both in normal operation and in the presence of faults occurring in sensors, actuators, or in the plant. Properties of interest range from stability (Gao and Antsaklis, 1991), disturbance attenuation (Niemann and Stoustrup, 2002), to model matching (Huang and Stengel, 1990), and optimality (Staroswiecki, 2003). FT mechanisms may be implemented either via pre-computed control laws or via on-line automatic redesign, using linear quadratic regulation (Looze *et al.*, 1985), pseudo-inverse (McLean and Aslam-Mir, 1991), model following (Gao and Antsaklis, 1992), eigenstructure assignment (Jiang, 1994), or model predictive control (Huzmezan and Maciejowski, 1998) techniques.

The Pseudo-Inverse Method (PIM) has first been introduced in flight control systems (Raza and Silverthorn, 1985), (Rattan, 1985). It is motivated by the need for pilots to keep faulty systems at hand, and it rests on adapting the control so that the faulty closed-loop system exhibits the same behavior as the nominal one (a model matching property). Since this is often impossible, the PIM finds the control that best approximates the nominal behavior. Unfortunately, this approach produces solutions whose closed-loop stability is not guaranteed. The stability problem was later addressed in (Gao and Antsaklis, 1991), based on the stability norm bounds developments (Yedavalli, 1985),

(Zhou and Khargonekar, 1987), under the name Modified Pseudo-Inverse Method (MPIM).

However, other drawbacks of the method are related to performance or robustness. They have been addressed in (Staroswiecki, 2005*a*), where it is proposed to search for the solution within a set of admissible models, instead of finding the best approximation of an ideal one. In this paper, the admissible model matching idea is developed for the fault tolerant control of the short-period longitudinal mode of a Boeing 747.

It is organized as follows : Section 2 presents the nominal versus the fault tolerant exact model matching problem. Section 3 addresses the case when exact model matching cannot be achieved : it presents the classical approximation of the exact problem (the pseudo-inverse and the modified pseudo-inverse approaches), and proposes the admissible model matching technique. Section 4 addresses the short-period mode control of the Boeing 747 aircraft longitudinal model, and compares the results obtained, by using the previous different approaches. Section 5 concludes the paper.

2. EXACT MODEL MATCHING

Let subscripts n and f respectively stand for nominal and faulty system conditions.

2.1 Nominal operation

The exact model matching technique with state-feedback aims at designing a control law :

$$u(t) = -K_n x(t) \tag{1}$$

for the system submitted to the constraint :

$$\dot{x}(t) = A_n x(t) + B_n u(t) \tag{2}$$

such that the resulting closed-loop behavior matches the reference model :

$$\dot{x}(t) = M^* x(t) \tag{3}$$

where $A_n \in R^{n\times n}$ and $B_n \in R^{n\times m}$ are the nominal system and control matrices, $x(t) \in R^n$ and $u(t) \in R^m$ are the state and control vectors, $M^* \in R^{n\times n}$ is a given stable matrix, and $K_n \in R^{m\times n}$ is the state-feedback matrix that is to be determined.

The solution of the exact model matching problem is obtained, by solving the equation :

$$A_n - B_n K_n = M^* \tag{4}$$

Assuming that matrix B_n is of full rank m, the necessary and sufficient condition for a solution to exist is :

$$Im\,(A_n - M^*) \subseteq Im\,(B_n) \tag{5}$$

and the solution is given by :

$$K_n = B_n^+ (A_n - M^*) \tag{6}$$

where notation $(.)^+$ stands for the left pseudo-inverse.

2.2 Faulty operation

When faults occur, assuming that a linear model is still able to capture its behavior, the system to be controlled is described by the pair (A_f, B_f). It follows that the FT model matching problem setting and solution are given by equations (1) to (6), with the only difference that index n (nominal) is replaced by index f (faulty).

3. APPROXIMATE VS. ADMISSIBLE MODEL MATCHING

Condition (5) is obviously satisfied from the system nominal design in the healthy situation, but its faulty counterpart will hold only for very particular faults, and therefore no solution will exist in most fault cases. For this reason, approximate rather than exact solutions may be of interest.

3.1 Approximate Model Matching

When exact model matching $\left(A_f - B_f K_f = M^*\right)$ is impossible, an approximate solution may be computed, by minimizing the criterion :

$$J(K_f) = \| \, (A_f - B_f K_f) - M^* \, \|_F^2 \tag{7}$$

where $\| \, . \, \|_F$ is the Frobenius matrix norm.

It is simple to show that the control law :

$$K_f^* = \arg\min J(K_f) \tag{8}$$

is still obtained, by using the faulty counterpart of (6), and that $J(K_f^*) = 0$ if and only if the compatibility condition in the faulty case holds.

Although approximate model matching seems reasonable, there are some drawbacks to the standard pseudo-inverse method (PIM), and its rationale is questionable :

1. The best Frobenius norm approximation of the stable matrix M^* may happen to be unstable. In the modified pseudo-inverse method (MPIM), the criterion $J(K_f)$ is minimized under the constraint that $A_f - B_f K_f$ is stable. However, the solution is based on sufficient stability conditions, and this may provide very conservative solutions, especially when the number of control inputs is significant. On another hand, it may be difficult to solve the constrained optimization problem in real-time, when solutions are urgently needed.

2. Even when it is stable, finding the system closest to the reference model does not guarantee that it will exhibit a satisfactory behavior, because no performance consideration is included in the problem setting. For example, the actuators may be overloaded or their mechanical limits may be forced, when implementing the PIM / MPIM solution.

3. The very meaning of "closest" is questionable, since the choice of the Frobenius norm-based distance is

arbitrary. When approximating the reference model, choosing another distance would provide another solution.

4. It follows from the problem setting that any fault (A_f, B_f) can be recovered, since there always exists a solution to the minimal distance problem. This point contradicts the fact that, in some fault situations, there is no control which can achieve fault recovery.

3.2 Admissible Model Matching

Admissible model matching (AMM) aims at giving the closed-loop system the property $A_f - B_f K_f \in \mathscr{M}$, where $\mathscr{M}$ is the problem specification, i.e. a set of closed loop matrices such that any solution of $\dot{x}(t) = Mx(t) : M \in \mathscr{M}$ is admissible in a sense that depends on the problem (stability, guaranteed performances, given constraints, etc.) (Staroswiecki, 2005*a*).

The set of admissible reference models $\mathscr{M}$ is defined off-line, and it can generally be described using inequality constraints :

$$\mathscr{M} = \{M \ \ s.t. \ \ \Phi(M) \leq 0\} \tag{9}$$

where the function $\Phi : R^{n\times n} \rightarrow R^d$ is known, and the vector inequality means that each component of $\Phi(M)$ is non-positive.

When a fault occurs, the problem becomes that of finding the control matrix K_f such that $\Phi(A_f - B_f K_f) \leq 0$. Note that, unlike the pseudo-inverse method, AMM guarantees that the behavior of the accommodated system will be admissible if this problem has a solution.

The necessary and sufficient condition for the admissible model matching problem to have a solution is that the faulty pair (A_f, B_f) satisfies :

$$\min_{K_f} \Phi(A_f - B_f K_f) \leq 0 \tag{10}$$

where (10) involves the optimization of the vector criterion $\Phi(M)$, and it means that the set of Pareto-optimal solutions intersects the negative cone in the space R^d (Staroswiecki, 2005*b*). It follows that the set of recoverable faults is the set of pairs (A_f, B_f) such that condition (10) holds true.

4. B747 LONGITUDINAL SHORT-PERIOD MODE CONTROL

In this section, the longitudinal model of the Boeing 747 aircraft is considered, with the aim of controlling the short-period oscillations. The aircraft flight condition is assumed to be a straight and constant level flight, at fixed altitude of 40000 ft, and Mach number of 0.8. Considering the CG (center of gravity) position to be $(0.25, 0, 0)$ of m.a.c. (mean aerodynamic chord), and the aircraft mass to be 19792 $slug$, the corresponding steady-state speed is 774 fps. The control δ_E (rad) is the elevator deflection. The exact, approximate, and admissible model matching approaches are applied to design a fault tolerant control that addresses the classical problem of loss of efficiency in the actuator. In addition, errors in the model of the faulty system are also introduced.

4.1 Nominal system

4.1.1. Open-loop model Let F_B be the aircraft body-axis reference frame. The aircraft longitudinal short-period dynamics is described by the state vector :

$$x = \begin{bmatrix} w & q \end{bmatrix}^T \tag{11}$$

where w (ft/sec) represents the inertial velocity in the F_B z-direction, and q (rad/sec) represents the pitch rate.

The linearized model of the system is :

$$\dot{x} = A_n x + B_n u \tag{12}$$

where the system and control matrices are given in (Etkin and Reid, 1996) :

$$A_n = \begin{bmatrix} -0.3127 & 774.0000 \\ -0.0010 & -0.4284 \end{bmatrix} \quad B_n = \begin{bmatrix} -17.9416 \\ -1.1577 \end{bmatrix} \tag{13}$$

In this flight configuration, matrix A_n has eigenvalues :

$$Sp\,(A_n) = \{-0.3706 \pm 0.8779\ i\} \tag{14}$$

The name "short-period" obviously follows from the aircraft natural mode having a period of 7.08 sec. It can also be observed that it is heavily damped, from the magnitude of the damping ratio which is 0.389.

4.1.2. Improving the nominal handling qualities by exact model matching The handling qualities refer to those characteristics of an aircraft that govern the ease and precision with which a pilot is able to perform his tasks (Etkin and Reid, 1996). They can be expressed in terms of the damping and frequency of the system mode, since these two parameters govern the system dynamic stability.

Although the B747 natural mode is inherently stable, the handling qualities of the system may be enhanced, by choosing an appropriate closed-loop reference model, and using an exact model matching controller design.

It can then be checked that the reference model :

$$M^* = \begin{bmatrix} -0.4573 & 731.9353 \\ -0.0103 & -3.1427 \end{bmatrix} \tag{15}$$

with eigenvalues :

$$Sp\,(M^*) = \{\lambda_1^*, \lambda_2^*\} = \{-1.8 \pm 2.4\ i\} \tag{16}$$

provides satisfactory performances, as far as longitudinal short-period oscillation is concerned. Indeed,

the reference model (15) has a damping coefficient of about 0.6 and a frequency of approximately 3 rad/sec, that the classical "thumb-print" criterion recommends as a good target for the aircraft short-period mode performance (O'Hara, 1967).

The state-feedback controller :

$$u = -K_n x \tag{17}$$

with :

$$K_n = \begin{bmatrix} -0.0081 & -2.3445 \end{bmatrix} \tag{18}$$

achieves exact model matching, and it is therefore chosen as the nominal system controller.

4.2 Faulty system

4.2.1. Considered faults The considered faults are the loss of efficiency of the actuator that controls the elevator deflection. Such faults are classically modeled by a multiplicative factor $\tau \in]0,1]$, so that the linear state-space representation of the faulty system will be described by the matrices $A_f = A_n$ and $B_f = (1-\tau)B_n$. Obviously $\tau = 0$ represents the nominal operation case, and $\tau = 1$ stands for the complete failure case, which is the situation where the actuator does not act on the system anymore.

The solution to the exact model matching problem is obviously obtained, by replacing the nominal pair (A_n, B_n) in (6) by the faulty pair (A_f, B_f). The control-gain matrix becomes :

$$K_f = \frac{1}{1-\tau} B_n^+ (A_n - M^*) \tag{19}$$

and then it follows that all faults are recoverable (except obviously for $\tau = 1$, since the system is no longer controlled at all in that case). Numerical values of K_f are given in Table 1 for some values of τ.

Table 1. Computing control matrix K_f based on K_n

τ	0.1	0.3	0.5	0.7	0.9
k_{f_1}	−0.0090	−0.0115	−0.0161	−0.0269	−0.0806
k_{f_2}	−2.6050	−3.3493	−4.6891	−7.8151	−23.4453

However, such a convenient case is far from being real. First, power limitations and actuator saturations have the consequence that the above solution works only for a limited range of values of τ. Second, the assumption that, after the fault, the system matrix is $A_f = A_n$ is not too realistic, since A_n results from the linearization of a non-linear behavior around an operating point that might well be changed by the fault occurrence.

In order to consider more complex situations, it is assumed that the fault is described by two coefficients, i.e. not only the control matrix $B_f = (1-\tau)B_n$, but also the system matrix $A_f = (1-\gamma)A_n$, where $\gamma \in (0, 0.1]$ is a given factor. One can check that the faulty counterpart of the consistency condition (5) does not hold anymore for $\gamma \neq 0$, therefore the faulty counterpart of relation (6) cannot be used for computing the faulty control matrix.

4.2.2. Pseudo-Inverse Method When the compatibility conditions are no longer satisfied, the pseudo-inverse method (PIM) looks for an approximate solution. Table 2 displays the pairs (undamped natural frequency, damping ratio) that are obtained for $\tau = 0.1$ and different values of γ, and it is easily seen that the behavior is very quickly degraded as γ increases.

Table 2. Evaluating system performance for PIM

γ	0.01	0.03	0.05	0.07
$k_{f_1}^{PIM}$	−0.0091	−0.0095	−0.0099	−0.0103
$k_{f_2}^{PIM}$	−2.1277	−1.1731	−0.2184	0.7363
ω^{PIM}	2.9850	2.9547	2.9240	2.8931
ζ^{PIM}	0.5190	0.3545	0.1867	0.0153

For example, for $\tau = 0.1$ and $\gamma = 0.1$, the pair (A_f, B_f) is :

$$A_f = \begin{bmatrix} -0.2814 & 696.6000 \\ -0.0009 & -0.3856 \end{bmatrix} \quad B_f = \begin{bmatrix} -16.1474 \\ -1.0419 \end{bmatrix} \tag{20}$$

The PIM approach provides the control-gain matrix :

$$K_{PIM} = \begin{bmatrix} -0.0109 & 2.1682 \end{bmatrix} \tag{21}$$

which results in the closed-loop system matrix :

$$M_{PIM} = \begin{bmatrix} -0.4572 & 731.6116 \\ -0.0122 & 1.8736 \end{bmatrix} \tag{22}$$

The poles being :

$$Sp(M_{PIM}) = \{0.7082 \pm 2.7565\, i\} \tag{23}$$

the PIM-based closed-loop is therefore unstable.

4.2.3. Modified Pseudo-Inverse Method Addressing the stability problem is the goal of the modified pseudo-inverse method (MPIM).

The best stable closed-loop matrix obtained with MPIM is :

$$M_{MPIM} = \begin{bmatrix} -0.2960 & 696.6146 \\ -0.0018 & -0.3846 \end{bmatrix} \tag{24}$$

and it is associated with the feedback :

$$K_{MPIM} = \begin{bmatrix} -0.0009 & -0.0009 \end{bmatrix} \tag{25}$$

Note that this approach indeed allows to stabilize the system, but it offers no real control of the behavior of the closed-loop system which results from fault accommodation, since one has :

$$Sp(M_{MPIM}) = \{-0.3403 \pm 1.1306\, i\} \tag{26}$$

which leads to a damping of about 0.3 and a frequency of approximately 1.2 rad/sec, that is not acceptable.

Table 3. Evaluating system performance for MPIM

τ	0.1	0.3	0.5	0.7	0.9
ω^{MPIM}	1.1807	1.1170	1.0494	0.9772	0.8992
ζ^{MPIM}	0.2882	0.3033	0.3214	0.3436	0.3717

Table 3 shows the evolution of the frequency and damping coefficient for $\gamma = 0.1$, and under increasing values of τ.

4.2.4. Admissible Model Matching The set of admissible closed-loop matrices is first specified in the admissible model matching (AMM), so as to account for closed-loop system performance.

Let M be a closed-loop matrix with eigenvalues $Sp(M) = \{\lambda_1, \lambda_2\}$. The undamped natural frequency and the damping coefficient are $\omega = \sqrt{\lambda_1 \lambda_2}$ and $\zeta = -\frac{\lambda_1+\lambda_2}{\sqrt{\lambda_1 \lambda_2}}$.

According to the "thumb-print" criterion, all closed-loop matrices with satisfactory short-period mode performance are described by :

$$\Phi_1(M) = \frac{(\zeta - 0.6)^2}{0.04} + \frac{(\omega - 3)^2}{0.64} - 1 \leq 0 \qquad (27)$$

which is taken as the first condition for admissibility.

Figure 1 depicts the set of acceptable pairs (ω, ζ). Moreover, it shows the unstable one provided by the PIM, and also the stable but unsatisfactory ones provided by the MPIM.

Note that defining admissibility via relation (27) is not very demanding. Indeed, if the pair (A_f, B_f) is still controllable, there is always a feedback law K_f such that $Sp(M) = Sp(M^*)$, and therefore $\Phi_1(M) = -1$ is obtained. It follows that specifying admissibility only by (27) boils down to designing fault accommodation on the basis of eigenvalues assignment, and in that case, any fault such that the system is still controllable can be recovered.

Note that, under this strategy, the discrepancies in the system trajectories can be easily computed by :

$$e(t) = V^* \left[T \exp(\Lambda^* t) T^{-1} - \exp(\Lambda^* t) \right] (V^*)^{-1} x_0 \qquad (28)$$

where $e(t) = x(t) - x^*(t)$, $x(t)$ is the solution of the accommodated closed-loop, $x^*(t)$ is the solution of the reference closed-loop, x_0 is the initial condition, $\Lambda^* = diag(\lambda_1^*, \lambda_2^*) = (V^*)^{-1} M^* V^* = (V)^{-1} MV$, and $T = (V^*)^{-1} V$. From equation (28) one has :

$$\|e(t)\| \leq \eta \left(1 + \|T\| \left\|T^{-1}\right\|\right) \|\exp(\Lambda^* t)\| \|x_0\| \qquad (29)$$

where $\eta = \|V^*\| \left\|(V^*)^{-1}\right\|$.

It follows that a second condition to define admissible closed-loops is naturally obtained, by imposing a given bound on the discrepancy by means of :

$$\Phi_2(M) = \|T\| \left\|T^{-1}\right\| - \alpha \leq 0 \qquad (30)$$

where α is a given constant. Note that defining the set of admissible closed-loops by $\Phi_1(M) \leq 0$ and

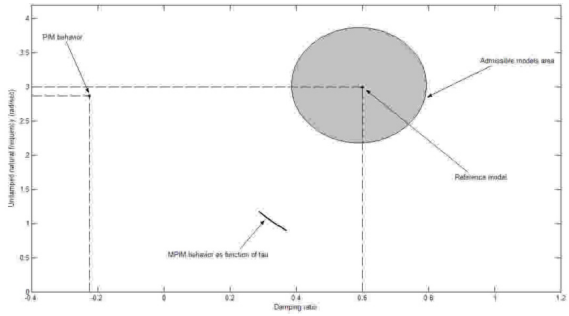

Fig. 1. Admissible Models Set

$\Phi_2(M) \leq 0$ establishes a clear link between recoverable faults and acceptable performances.

Let, for example, the bound on the distance between the system trajectories be defined by $\alpha = 1.10$. According to the admissible model matching development, the fault defined by the pair $(\gamma, \tau) = (0.05, 0.5)$ will give the system and control matrices $A_f = 0.95 A_n$ and $B_f = 0.5 B_n$:

$$A_f = \begin{bmatrix} -0.2971 & 735.3000 \\ -0.0010 & -0.4070 \end{bmatrix} \quad B_f = \begin{bmatrix} -8.9708 \\ -0.5788 \end{bmatrix} \qquad (31)$$

The control matrix :

$$K_{AMM_1} = \begin{bmatrix} -0.0173 & -4.7356 \end{bmatrix} \qquad (32)$$

achieves the closed-loop behavior :

$$M_{AMM_1} = \begin{bmatrix} -0.4518 & 692.8183 \\ -0.0109 & -3.1482 \end{bmatrix} \qquad (33)$$

whose poles :

$$Sp(M_{AMM_1}) = \{-1.8 \pm 2.4\, i\} \qquad (34)$$

are obviously the reference ones, so that the dynamic performance parameters $(\omega^{AMM_1} = 3.0, \zeta^{AMM_1} = 0.6)$ satisfy the two conditions $\Phi_1(M_{AMM_1}) = -1 \leq 0$ and $\Phi_2(M_{AMM_1}) = -0.04 \leq 0$. Therefore, the fault $(\gamma = 0.05, \tau = 0.5)$ is recoverable.

On the contrary, the fault defined by the pair $(\gamma, \tau) = (0.1, 0.5)$ will give the system and control matrices $A_f = 0.9 A_n$ and $B_f = 0.5 B_n$:

$$A_f = \begin{bmatrix} -0.2814 & 696.6000 \\ -0.0009 & -0.3856 \end{bmatrix} \quad B_f = \begin{bmatrix} -8.9708 \\ -0.5788 \end{bmatrix} \qquad (35)$$

In this case, the control matrix :

$$K_{AMM_2} = \begin{bmatrix} -0.0185 & -4.7802 \end{bmatrix} \qquad (36)$$

achieves the closed-loop behavior :

$$M_{AMM_2} = \begin{bmatrix} -0.4474 & 653.7175 \\ -0.0116 & -3.1526 \end{bmatrix} \qquad (37)$$

with poles :

$$Sp(M_{AMM_2}) = Sp(M_{AMM_1}) \qquad (38)$$

However, even the dynamic parameters pair $(\omega^{AMM_2} = \omega^{AMM_1}, \zeta^{AMM_2} = \zeta^{AMM_1})$ stands for the first condition, $\Phi_1(M_{AMM_2}) = -1 \leq 0$, it does not hold for the second one, since $\Phi_2(M_{AMM_2}) = 0.04 \not\leq 0$, and therefore the fault $(\gamma, \tau) = (0.1, 0.5)$ is a non-recoverable one.

5. CONCLUSION

Fault tolerance (FT) represents the ability of a controlled system to still achieve the control objectives, despite the occurrence of faults, maybe at the cost of limited performance degradation. The classical and the modified pseudo-inverse methods are "standards", when model matching is the control objective. However, they exhibit some drawbacks that can be overcome, by using the new point of view proposed in this paper : using a set of admissible models, rather than searching for an optimal approximate one, will indeed provide a guarantee about the faulty system behavior, expressed in terms of stability or closed-loop performance.

The approach has been illustrated by the longitudinal short-period mode control of a Boeing 747 aircraft, an application where the model matching objective addresses the handling qualities requirements, that are synthesized by the iso-rating curves, which specify the aircraft flight regime accepted performances. The considered faults are modeled through the loss of efficiency of the actuator that controls the elevator deflection, and an extra factor that complicates the problem has been introduced to represent estimation errors in the system matrix.

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